



**E0452**

Place student sticker here

**Note:**

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

## Diskrete Wahrscheinlichkeitstheorie

**Exam:** / Endterm **Date:** Friday 1<sup>st</sup> August, 2025  
**Examiner:** **Time:** 08:00 – 10:00

|   | P 1 | P 2 | P 3 | P 4 | P 5 |
|---|-----|-----|-----|-----|-----|
| I |     |     |     |     |     |

### Working instructions

- This exam consists of **12 pages** with a total of **5 problems**.  
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 51 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **single sheet of A4 paper** with handwritten notes on both sides
  - one **analog dictionary** English ↔ native language **without notes**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.
- Place your student ID on your desk.

Left room from \_\_\_\_\_ to \_\_\_\_\_ / Early submission at \_\_\_\_\_



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## Problem 1 True or false (12 credits)

Decide whether the statements are true or false (clearly indicate your answer) and provide a justification (proof, counterexample, or explanation via known results).

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a) Let  $(\Omega, \mathcal{A}, P)$  be a probability space and  $A, B \in \mathcal{A}$  with  $P(A) > 0$  and  $P(B) > 0$ . Then  $P(A | B) = P(B | A)$ .

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b) Let  $(\Omega, \mathcal{A}, P_\theta)$  be a statistical model and let  $\hat{\theta}_1, \hat{\theta}_2$  be unbiased estimators of  $\theta$  in  $P_\theta$ . Then  $\frac{1}{3}\hat{\theta}_1 + \frac{2}{3}\hat{\theta}_2$  is an unbiased estimator of  $\theta$  as well.





c) Upon receiving a large batch of switches, you take a subsample (uniformly at random) of size  $n$ . You run a (suitable) hypothesis test at significance level  $\alpha = 0.05$  to test  $H_0$ : “the proportion of defective switches in the batch is at most what the manufacturer claimed” versus  $H_1$ : “the proportion of defective switches is larger.” The test rejects  $H_0$  at level  $\alpha = 0.05$ . You can therefore conclude that the probability that the entire batch conforms to the claimed proportion is at most 5%.

d) Let  $X, Y$  be iid following  $\text{Exp}(\lambda)$  for  $\lambda > 0$ . Then the cdf of the ratio  $X/Y$  is

$$F_{X/Y}(t) = \begin{cases} \frac{t}{1+t} & \text{if } t > 0, \\ 0 & \text{otherwise.} \end{cases}$$

**Hint:** The pdf of the exponential distribution is  $p_{\text{Exp}(\lambda)}(x) = \lambda e^{-\lambda x}$  for  $x \geq 0$  (and 0 otherwise). Note that  $F_{X/Y}(t) = P(X < t \cdot Y)$ .





## Problem 2 Definitions (6 credits)

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a) Write out the central limit theorem with all required assumptions.

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b) Give the full general definition of the type of convergence in the central limit theorem.





### Problem 3 Density computations (9 credits)

Let  $X$  be a continuous RVRV with probability density function

$$p_X(x) = \begin{cases} \frac{x}{c} & \text{for } x \in [0, 2], \\ 0 & \text{otherwise,} \end{cases}$$

for some  $c \in \mathbb{R}$ .

a) Find  $c \in \mathbb{R}$  such that  $p_X$  is a valid pdf.



b) Determine the cumulative distribution function (cdf)  $F_X(x)$  of  $X$ .





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c) Calculate the mean and median of  $X$ .

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d) Let  $Y = \ln(X)$ . Determine the pdf of the random variable  $Y$ .

**Hint:** You may use the change of variables formula for densities:  $p_Y(y) = p_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$ .



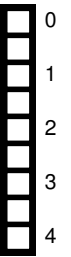


## Problem 4 Estimating and testing (12 credits)

Let  $X_1, \dots, X_n$  be iid RVRVs from a Poisson distribution  $\text{Poi}(\lambda)$  with unknown rate parameter  $\lambda > 0$ .

a) Explicitly derive the maximum likelihood estimator (mle)  $\hat{\lambda}_{\text{mle}}$  for  $\lambda$ .

**Hint:** The pmf of a  $\text{Poi}(\lambda)$  distributed random variable is  $p_{\text{Poi}(\lambda)}(x) : \mathbb{N} \rightarrow [0, 1], x \mapsto \frac{e^{-\lambda} \lambda^x}{x!}$ , which has mean and variance  $\lambda$ . Recall that the derivative of  $\ln(t)$  is  $1/t$  for  $t > 0$ .



b) Is  $\hat{\lambda}_{\text{mle}}$  an unbiased estimator for  $\lambda$ ?





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c) Use the central limit theorem (clt) to find the approximate distribution of  $\hat{\lambda}_{\text{mle}}$  for large  $n$ .

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d) In a light bulb factory, the number of defectives per batch is assumed to follow a Poisson distribution. Historically, the average number of defects per batch has been  $\lambda_0 = 4$ . After a change in the manufacturing process, you test  $n = 100$  new batches and find an average of  $\bar{x}_{100} = 5$  defects.

Construct an *approximate* hypothesis test to test whether the average number of defects per batch has *increased above* the historical average at a significance level of  $\alpha = 0.05$ . Based on this test, should you reject the null hypothesis?

**Hint:** You may use that the 0.95-quantile of a standard normal distribution is  $z_{0.05} \approx 1.64$ .





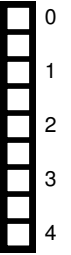


## Problem 5 Duck hunt (12 credits)

Let  $m, n \in \mathbb{N}_{>0}$ . A group of  $n$  hunters with perfect aim simultaneously shoot at  $m$  ducks, where each hunter selects its victim arbitrarily (that is, uniformly at random) and independent of the other hunters. So it can happen that a duck gets selected by multiple hunters. Let  $X$  be the  $\mathbb{N}$ -valued random variable modeling the number of surviving ducks after all hunters have pulled the trigger (each of them only shoots once).

a) Calculate the expectation  $\mathbb{E}[X]$ .

**Hint:** Enumerate the ducks and define the event  $A_i :=$  “the  $i$ -th duck survives.” Then construct the random variable  $X$  using  $\chi_{A_i}$ .





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b) Calculate the variance  $\text{var}[X]$ .



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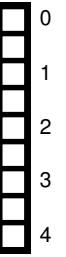


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c) Assume now there are 2 hunters. Find  $k \in \mathbb{N}$  as a function of  $m$  such that the number of surviving ducks is within  $[\mathbb{E}[X] - k, \mathbb{E}[X] + k]$  with a probability of at least 90%.

**Hint:** Rewrite this condition so you can use the Chebyshev inequality.



